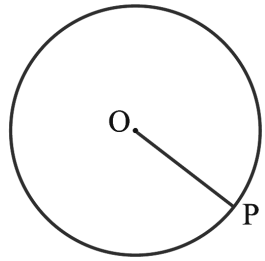
CLASS X : CHAPTER - 10
CIRCLES

IMPORTANT FORMULAS & CONCEPTS

Circle

The collection of all the points in a plane, which are at a fixed distance from a fixed point in the plane, is called a circle.

- The fixed point is called the *centre* of the circle and the fixed distance is called the *radius* of the circle. In the below figure, O is the centre and the length OP is the radius of the circle.



- The line segment joining the centre and any point on the circle is also called a *radius* of the circle.

- A circle divides the plane on which it lies into three parts. They are: (i) inside the circle, which is also called the *interior* of the circle; (ii) the *circle* and (iii) outside the circle, which is also called the *exterior* of the circle. The circle and its interior make up the *circular region.*

- The chord is the line segment having its two end points lying on the circumference of the circle.

- The chord, which passes through the centre of the circle, is called a *diameter* of the circle.

- A *diameter is the longest chord and all diameters have the same length, which is equal to two times the radius.*

- A piece of a circle between two points is called an *arc*.

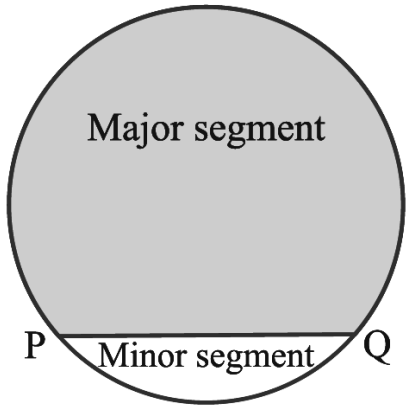
- The longer one is called the *major arc* PQ and the shorter one is called the *minor arc* PQ.

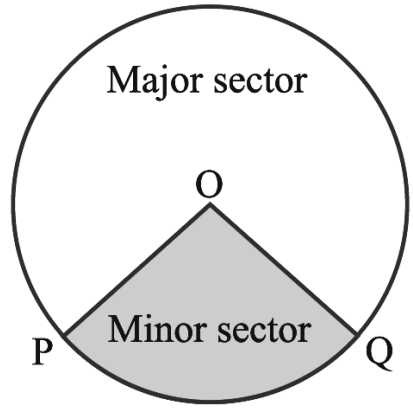
- The length of the complete circle is called its *circumference.*

- The region between a chord and either of its arcs is called a *segment* of the circular region or simply a *segment* of the circle. There are two types of segments also, which are the *major segment* and the *minor segment*.

- The region between an arc and the two radii, joining the centre to the end points of the arc is called a *sector*. The minor arc corresponds to the *minor sector* and the major arc corresponds to the *major sector*.

- In the below figure, the region OPQ is the minor sector and remaining part of the circular region is the major sector. When two arcs are equal, that is, each is a semicircle, then both segments and both sectors become the same and each is known as a *semicircular region.*





Points to Remember :

- A circle is a collection of all the points in a plane, which are equidistant from a fixed point in the plane.
- Equal chords of a circle (or of congruent circles) subtend equal angles at the centre.
- If the angles subtended by two chords of a circle (or of congruent circles) at the centre (corresponding centre) are equal, the chords are equal.
- The perpendicular from the centre of a circle to a chord bisects the chord.
- The line drawn through the centre of a circle to bisect a chord is perpendicular to the chord.
- There is one and only one circle passing through three non-collinear points.
- Equal chords of a circle (or of congruent circles) are equidistant from the centre (or corresponding centres).
- Chords equidistant from the centre (or corresponding centres) of a circle (or of congruent circles) are equal.
- If two arcs of a circle are congruent, then their corresponding chords are equal and conversely, if two chords of a circle are equal, then their corresponding arcs (minor, major) are congruent.
- Congruent arcs of a circle subtend equal angles at the centre.
- The angle subtended by an arc at the centre is double the angle subtended by it at any point on the remaining part of the circle.
- Angles in the same segment of a circle are equal.\
- Angle in a semicircle is a right angle.
- If a line segment joining two points subtends equal angles at two other points lying on the same side of the line containing the line segment, the four points lie on a circle.
- The sum of either pair of opposite angles of a cyclic quadrilateral is 180°.
- If the sum of a pair of opposite angles of a quadrilateral is 180°, then the quadrilateral is cyclic.

Secant to a Circle

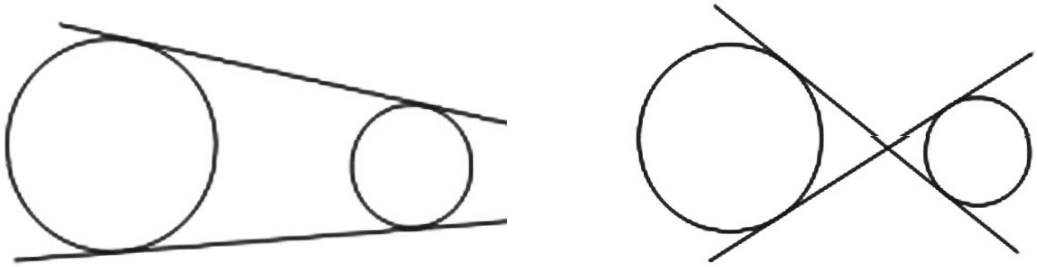
A secant to a circle is a line that intersects the circle at exactly two points.

Tangent to a Circle

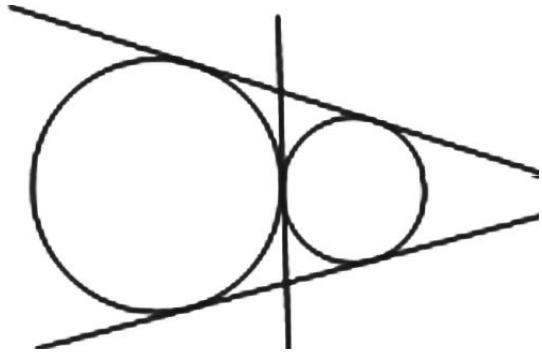
A tangent to a circle is a line that intersects the circle at only one point.

Given two circles, there are lines that are tangents to both of them at the same time.

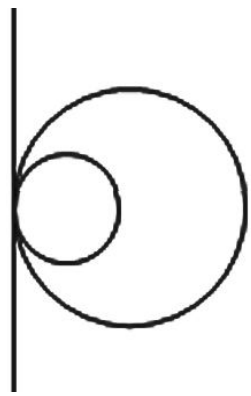
☞ If the circles are separate (do not intersect), there are four possible common tangents:



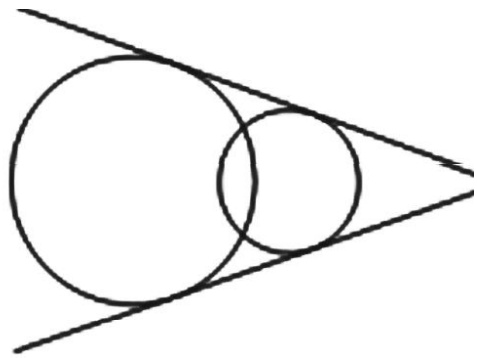
☞ If the two circles touch at just one point, there are three possible tangent lines that are common to both:



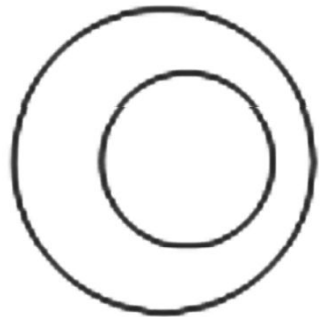
☞ If the two circles touch at just one point, with one inside the other, there is just one line that is a tangent to both:



☞ If the circles overlap - i.e. intersect at two points, there are two tangents that are common to both:



☞ If the circles lie one inside the other, there are no tangents that are common to both. A tangent to the inner circle would be a secant of the outer circle.



- The tangent to a circle is perpendicular to the radius through the point of contact.
- *The lengths of tangents drawn from an external point to a circle are equal.*
- The centre lies on the bisector of the angle between the two tangents.
- "If a line in the plane of a circle is perpendicular to the radius at its endpoint on the circle, then the line is tangent to the circle".

MCQ WORKSHEET-I
CLASS X: CHAPTER – 10
CIRCLES

1. Find the length of tangent drawn to a circle with radius 7 cm from a point 25 cm away from the centre.

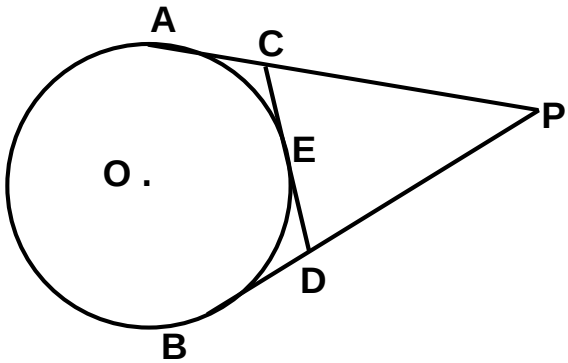
(a) 24 cm (b) 27 cm (c) 26 cm (d) 25 cm

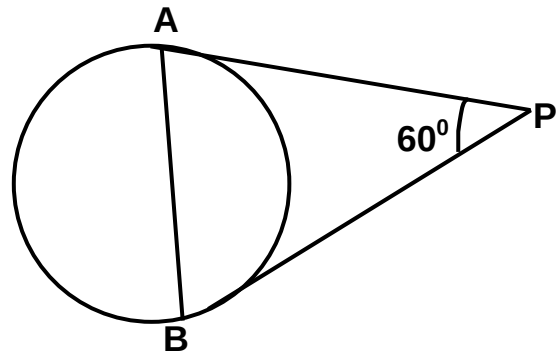
2. A point P is 26 cm away from the centre of a circle and the length of the tangent drawn from P to the circle is 24 cm. Find the radius of the circle.

(a) 11 cm (b) 10 cm (c) 16 cm (d) 15 cm

3. From an external point P, tangents PA and PB are drawn to a circle with centre O. If CD is the tangent to the circle at a point E and PA = 14 cm, find the perimeter of the ΔPCD.

(a) 28 cm (b) 27 cm (c) 26 cm (d) 25 cm



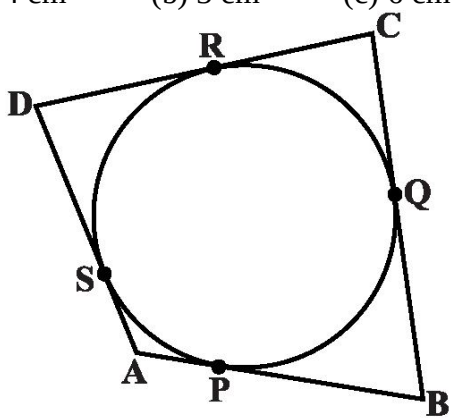


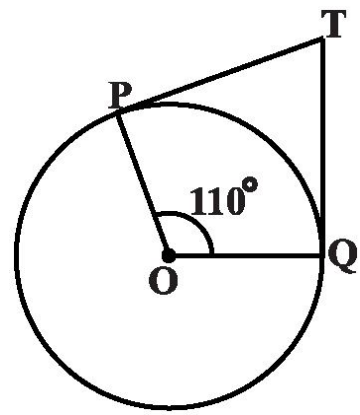
4. In the above sided figure, PA and PB are tangents such that PA = 9cm and ∠APB = 60^0. Find the length of the chord AB.

(a) 4 cm (b) 7 cm (c) 6 cm (d) 9 cm

5. In the below figure the circle touches all the sides of a quadrilateral ABCD whose three sides are AB = 6 cm, BC = 7 cm, CD = 4 cm. Find AD.

(a) 4 cm (b) 3 cm (c) 6 cm (d) 9 cm





6. In the above sided Fig., if TP and TQ are the two tangents to a circle with centre O so that ∠POQ = 110°, then ∠PTQ is equal to

(a) 60^0 (b) 70^0 (c) 80^0 (d) 90^0

7. If tangents PA and PB from a point P to a circle with centre O are inclined to each other at angle of 80°, then ∠POA is equal to

(a) 60^0 (b) 70^0 (c) 80^0 (d) 50^0

8. The length of a tangent from a point A at distance 5 cm from the centre of the circle is 4 cm. Find the radius of the circle.

(a) 4 cm (b) 3 cm (c) 6 cm (d) 5 cm

9. From a point P, 10 cm away from the centre of a circle, a tangent PT of length 8 cm is drawn. Find the radius of the circle.

(a) 4 cm (b) 7 cm (c) 6 cm (d) 5 cm

10. PT is tangent to a circle with centre O, OT = 56 cm, TP = 90 cm, find OP

(a) 104 cm (b) 107 cm (c) 106 cm (d) 105 cm

11. TP and TQ are the two tangents to a circle with center O so that angle $\angle POQ = 130^0$. Find $\angle PTQ$.

(a) 50^0 (b) 70^0 (c) 80^0 (d) none of these

12. From a point Q, the length of the tangent to a circle is 40 cm and the distance of Q from the centre is 41 cm. Find the radius of the circle.

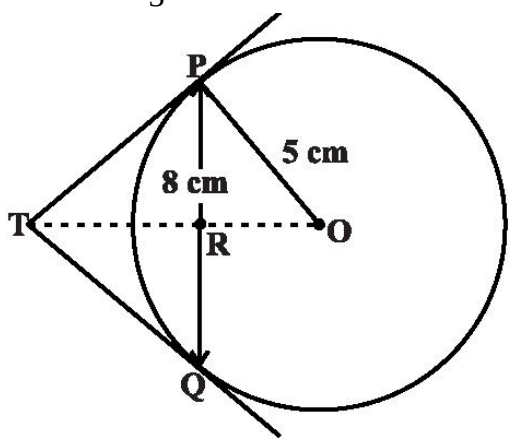
(a) 4 cm (b) 3 cm (c) 6 cm (d) 9 cm

13. The common point of a tangent to a circle with the circle is called _____

(a) centre (b) point of contact (c) end point (d) none of these.

14. PQ is a chord of length 8 cm of a circle of radius 5 cm. The tangents at P and Q intersect at a point T (see below figure). Find the length TP.

(a) $\frac{20}{3}$ cm (b) $\frac{10}{3}$ cm (c) $\frac{40}{3}$ cm (d) none of these



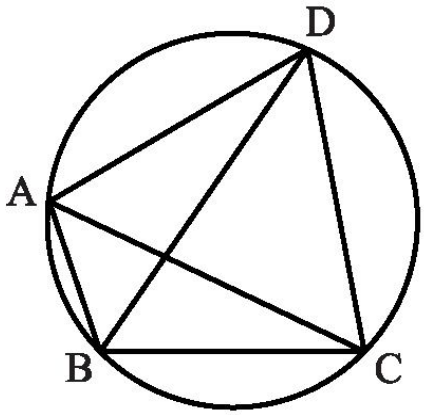
15. *The lengths of tangents drawn from an external point to a circle are equal.*

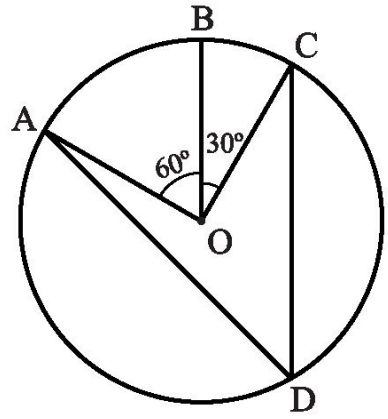
(a) half (b) one third (c) one fourth (d) equal

MCQ WORKSHEET-II
CLASS X: CHAPTER – 10
CIRCLES

1. In below Fig, ABCD is a cyclic quadrilateral in which AC and BD are its diagonals. If ∠ DBC = 55° and ∠BAC = 45°, find ∠BCD.

(a) 80º (b) 60º (c) 90º (d) none of these





2. In above sided Fig, A,B and C are three points on a circle with centre O such that ∠BOC = 30° and ∠AOB = 60°. If D is a point on the circle other than the arc ABC, find ∠ADC.

(a) 45º (b) 60º (c) 90º (d) none of these

3. A chord of a circle is equal to the radius of the circle. Find the angle subtended by the chord at a point on the minor arc

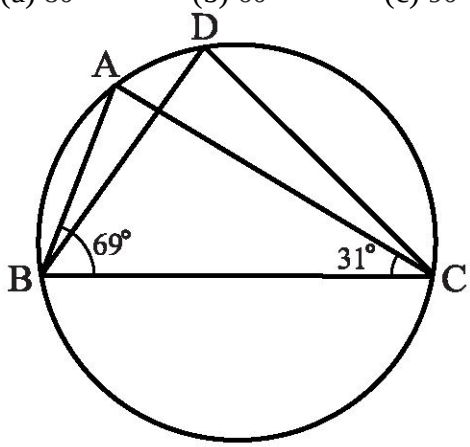
(a) 150^0 (b) 30º (c) 60º (d) none of these

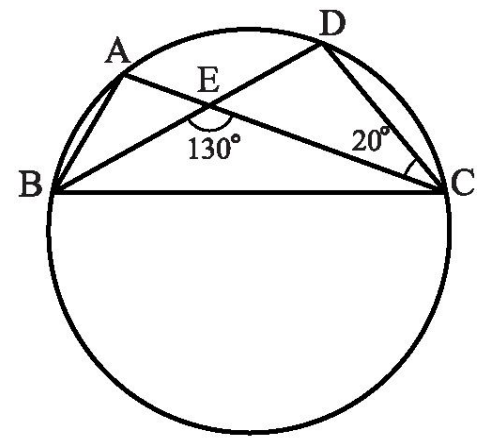
4. A chord of a circle is equal to the radius of the circle. Find the angle subtended by the chord at a point on the major arc.

(a) 150^0 (b) 30º (c) 60º (d) none of these

5. In the below Fig., ∠ABC = 69°, ∠ACB = 31°, find ∠BDC.

(a) 80º (b) 60º (c) 90º (d) 100^0





6. In the above sided Fig., A, B, C and D are four points on a circle. AC and BD intersect at a point E such that ∠BEC = 130° and ∠ECD = 20°. Find ∠BAC.

(a) 110º (b) 150º (c) 90º (d) 100^0

7. ABCD is a cyclic quadrilateral whose diagonals intersect at a point E. If ∠DBC = 70°, ∠BAC is 30°, find ∠BCD.

(a) 80º (b) 60º (c) 90º (d) 100^0

8. ABCD is a cyclic quadrilateral. If ∠BCD = 100°, ∠ABD is 30°, find ∠ABD.

(a) 80º (b) 60º (c) 90º (d) 70^0

9. ABCD is a cyclic quadrilateral. If ∠DBC= 80°, ∠BAC is 40°, find ∠BCD.

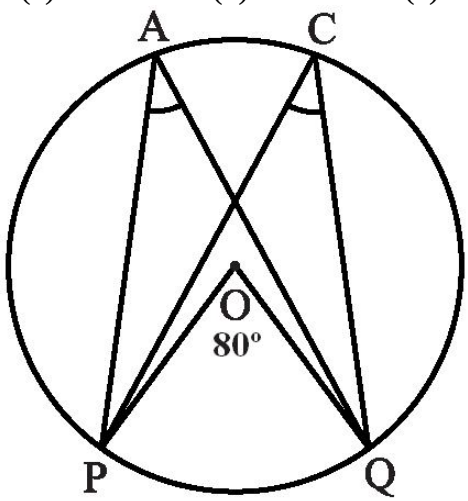
(a) 80º (b) 60º (c) 90º (d) 70^0

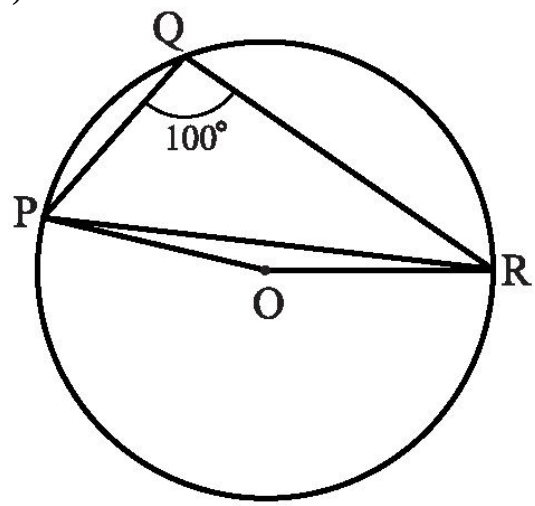
10. ABCD is a cyclic quadrilateral in which BC is parallel to AD, ∠ADC = 110^0 and ∠BAC = 50^0. Find ∠DAC

(a) 80º (b) 60º (c) 90º (d) 170^0

11. In the below figure, ∠POQ= 80^0, find ∠PAQ

(a) 80º (b) 40º (c) 100º (d) none of these





12. In the above figure, ∠PQR = 100°, where P, Q and R are points on a circle with centre O. Find ∠OPR.

(a) 80º (b) 40º (c) 10º (d) none of these

MCQ WORKSHEET-III
CLASS X: CHAPTER – 10
CIRCLES

1. Distance of chord AB from the centre is 12 cm and length of the chord is 10 cm. Then diameter of the circle is

A. 26 cm B. 13 cm C. $\sqrt{244}$ cm D. 20 cm

2. Two circles are drawn with side AB and AC of a triangle ABC as diameters. Circles intersect at a point D, Then

A. $\angle$ADB and $\angle$ADC are equal B. $\angle$ADB and $\angle$ADC are compementary

C. Points B, D, C are collinear D. none of these

3. The region between a chord and either of the arcs is called

A. an arc B. a sector C. a segment D. a semicircle

4. A circle divides the plane in which it lies, including circle in

A. 2 parts B. 3 parts C. 4 parts D. 5 parts

5. If diagonals of a cyclic quadrilateral are the diameters of a circle through the vertices of a quadrilateral, then quadrilateral is a

A. parallelogram B. square C. rectangle D. trapezium

6. Given three non collinear points, then the number of circles which can be drawn through these three points are

A. one B. zero C. two D. infinite

7. In a circle with centre O, AB and CD are two diameters perpendicular to each other. The length of chord AC is

A. 2 AB B. $\sqrt{2}$ AB C. $\frac{1}{2}$AB D. $\frac{1}{\sqrt{2}}$AB

8. If AB is a chord of a circle, P and Q are the two points on the circle different from A and B, then

A. $\angle APB = \angle AQB$

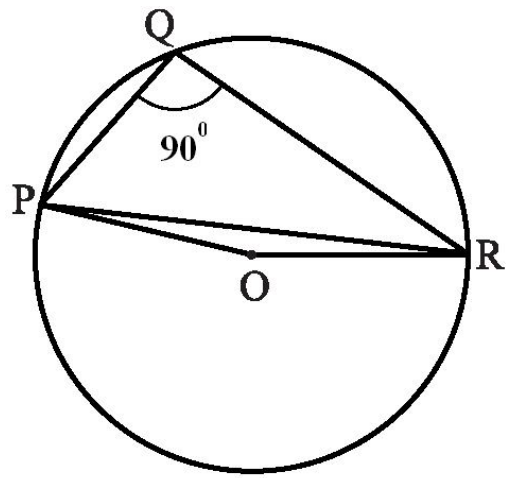
B. $\angle APB + \angle AQB = 180^\circ$

C. $\angle APB + \angle AQB = 90^\circ$

D. $\angle APB + \angle AQB = 180^\circ$

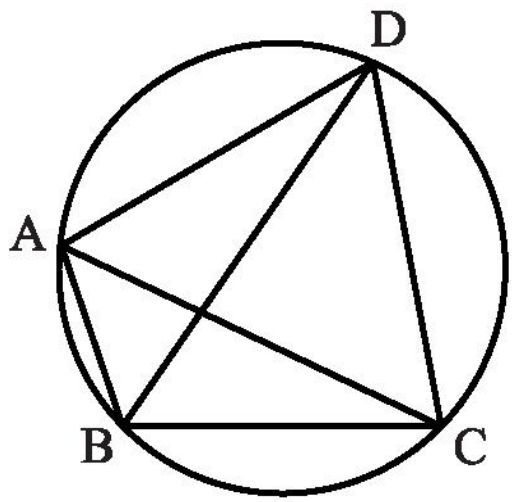
9. In the above figure, ∠PQR = 90°, where P, Q and R are points on a circle with centre O. Find reflex∠POR.

(a) 180º (b) 140º (c) 45º (d) none of these



10. In below Fig, ABCD is a cyclic quadrilateral in which AC and BD are its diagonals. If ∠ DBC = 60° and ∠BAC = 30°, find ∠BCD.

(a) 80º (b) 60º (c) 90º (d) none of these



PRACTICE QUESTIONS
CLASS X: CHAPTER – 10
CIRCLES

1. Prove that "The tangent at any point of a circle is perpendicular to the radius through the point of contact".

2. Prove that "The lengths of tangents drawn from an external point to a circle are equal."

3. Prove that "The centre lies on the bisector of the angle between the two tangents drawn from an external point to a circle."

4. Find the length of the tangent drawn to a circle of radius 3 cm, from a point distant 5 cm from the centre.

5. A point P is at a distance 13 cm from the centre C of a circle and PT is a tangent to the given circle. If PT = 12 cm, find the radius of the circle.

6. From appoint Q, the length of the tangent to a circle is 24 cm and the distance of Q from the centre of the circle is 25 cm. Find the radius of the circle.

7. The tangent to a circle of radius 6 cm from an external point P, is of length 8 cm. Calculate the distance of P from the nearest point of the circle.

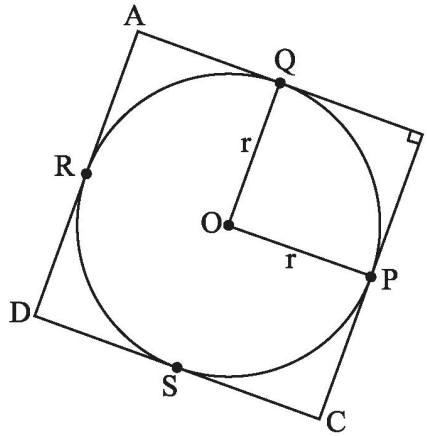
8. Prove that in two concentric circles, the chord of the bigger circle, which touches the smaller circle is bisected at the point of contact.

9. ΔPQR circumscribes a circle of radius r such that angle Q = 90^0, PQ = 3 cm and QR = 4 cm. Find r.

10. Prove that the parallelogram circumscribing a circle is a rhombus.
OR
If all the sides of a parallelogram touch the circle, show that the parallelogram is a rhombus.

11. ABC is an isosceles triangle in which AB = AC, circumscribed about a circle. Show that BC is bisected at the point of contact.

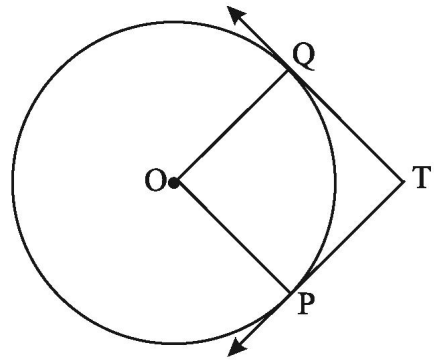
12. In Fig., a circle is inscribed in a quadrilateral ABCD in which $\angle B = 90^0$. If AD = 23 cm, AB = 29 cm and DS = 5 cm, find the radius (r) of the circle.



13. ABCD is a quadrilateral such that $\angle D = 90^0$. A circle C(O, r) touches the sides AB, BC, CD and DA at P, Q, R and S respectively. If BC = 38 cm, CD = 25 cm and BP = 27 cm, find r.

14. An isosceles triangle ABC is inscribed in a circle. If AB = AC = 13 cm and BC = 10 cm, find the radius of the circle.

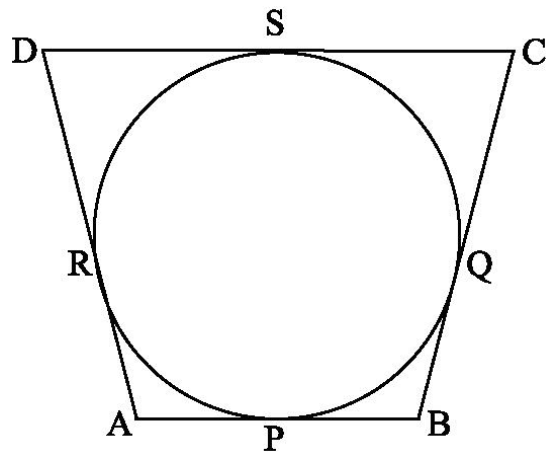
15. Two tangents TP and TQ are drawn from a external point T to a circle with centre O, as shown in fig. If they are inclined to each other at an angle of 100^0 then what is the value of $\angle POQ$?



16. The incircle of ΔABC touches the sides BC, CA and AB at D, E and F respectively. If AB = AC, prove that BD = CD.

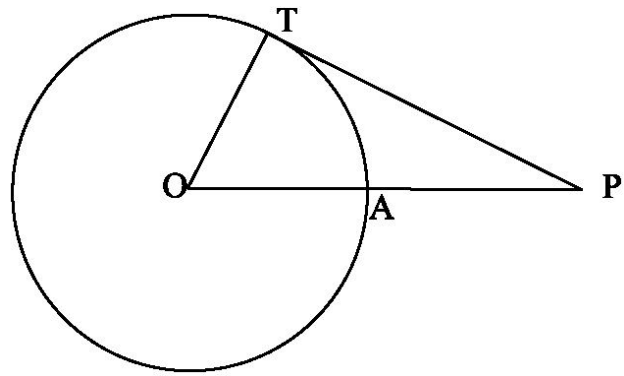
17. XP and XQ are tangents from X to the circle with O, R is a point on the circle and a tangent through R intersect XP and XQ at A and B respectively. Prove that XA + AR = XB + BR.

18. A circle touches all the four sides of a quadrilateral ABCD with AB = 6 cm, BC = 7cm and CD = 4 cm. Find AD.

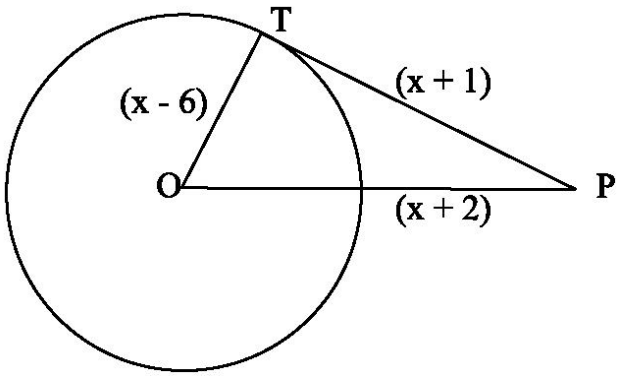


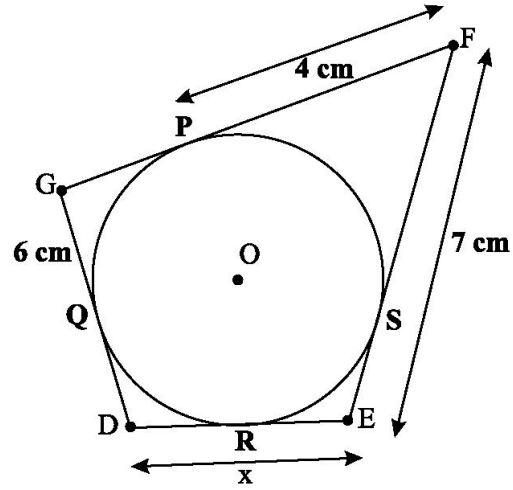
19. TP and TQ are tangents to a circle with centre O at P and Q respectively. PQ = 8cm and radius of circle is 5 cm. Find TP and TQ.

20. In the below figure PT is tangent to a circle with centre O, PT = 36 cm, AP = 24 cm. Find the radius of the circle.



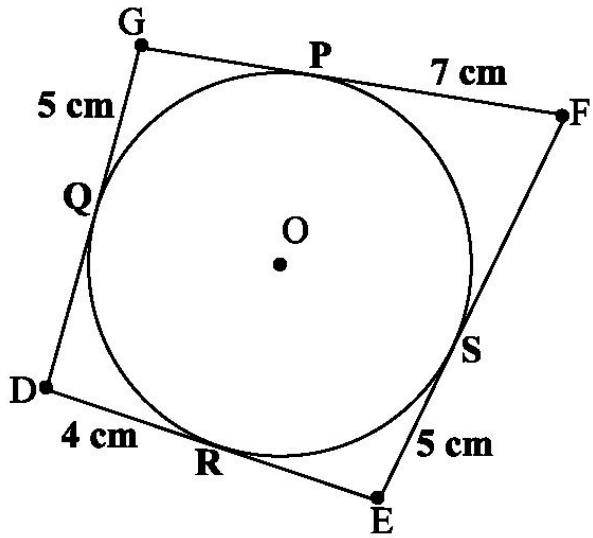
21. In the below figure, find the actual length of sides of ΔOTP.





22. In the above sided figure, find the value of x.

23. Find the perimeter of DEFG.



24. Two tangents TP and TQ are drawn to a circle with centre O from an external point T. Prove that ∠PTQ = 2∠OPQ.

25. PQ is a chord of length 8 cm of a circle of radius 5 cm. The tangents at P and Q intersect at a point T. Find the length TP.

26. Prove that the perpendicular at the point of contact to the tangent to a circle passes through the centre.

27. The length of a tangent from a point A at distance 5 cm from the centre of the circle is 4 cm. Find the radius of the circle.

28. Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.

29. A quadrilateral ABCD is drawn to circumscribe a circle. Prove that AB + CD = AD + BC

30. Prove that the angle between the two tangents drawn from an external point to a circle is supplementary to the angle subtended by the line-segment joining the points of contact at the centre.

31. Prove that opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre of the circle.

OR

A circle touches all the four sides a quadrilateral ABCD. Prove that the angles subtended at the centre of the circle by the opposite sides are supplementary.

32. PA and PB are the two tangents to a circle with centre O in which OP is equal to the diameter of the circle. Prove that APB is an equilateral triangle.

33. Prove that the intercept of a tangent between two parallel tangents to a circle subtends a right angle at the center of the circle.

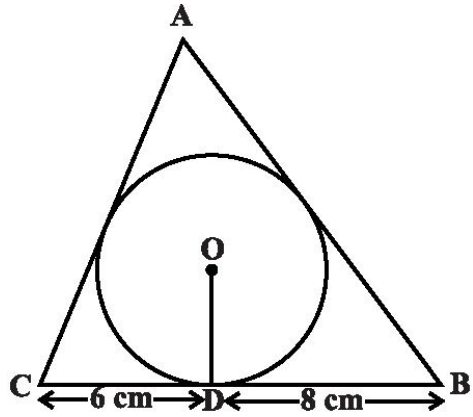
34. If PQ and RS are two parallel tangents to a circle with centre O and another tangent X, with point of contact C intersects PQ at A and RS at B. Prove that $\angle AOB = 90^o$.

35. The incircle of ΔABC touches the sides BC, CA and AB at D, E and F respectively. If AB = AC, prove that BD = DC.

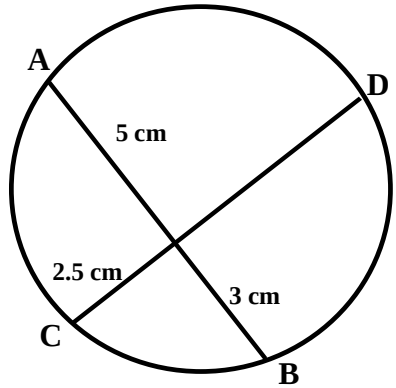
36. Two tangents PA and PB are drawn to the circle with center O, such that $\angle APB = 120^0$. Prove that OP = 2AP.

37. A circle is touching the side BC of ΔABC at P and is touching AB and AC when produced at Q and R respectively. Prove that AQ = ½ (Perimeter of Δ ABC).

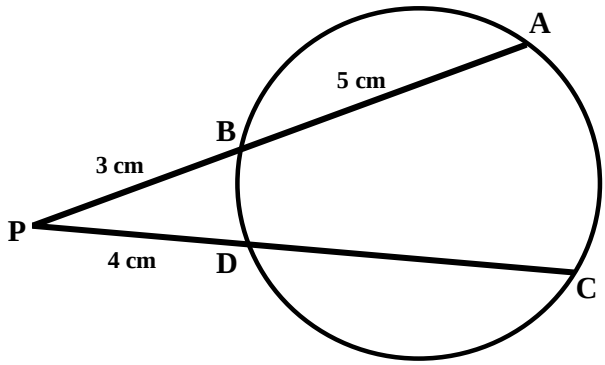
38. A triangle ABC is drawn to circumscribe a circle of radius 4 cm such that the segments BD and DC into which BC is divided by the point of contact D are of lengths 8 cm and 6 cm respectively. Find the sides AB and AC.



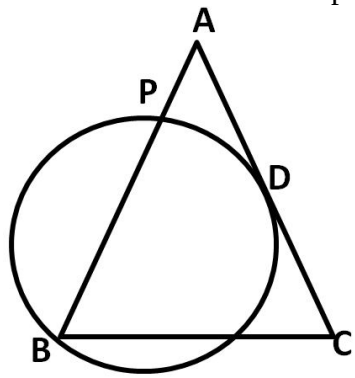
39. In figure, chords AB and CD of the circle intersect at O. OA = 5cm, OB = 3cm and OC = 2.5cm. Find OD.



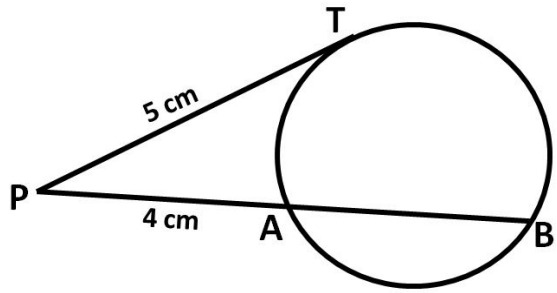
40. In figure. Chords AB and CD intersect at P. If AB = 5cm, PB = 3cm and PD = 4cm. Find the length of CD.



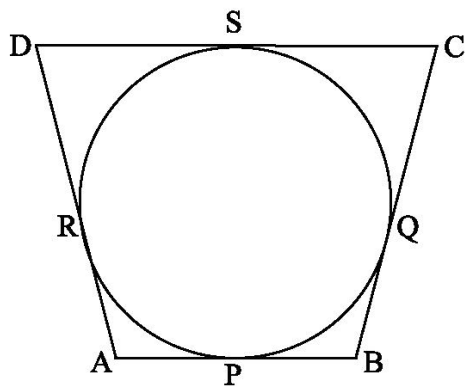
41. In the figure, ABC is an isosceles triangle in which AB = AC. A circle through B touches the side AC at D and intersect the side AB at P. If D is the midpoint of side AC, Then AB = 4AP.



42. In the figure. Find the value of AB Where PT = 5cm and PA = 4cm.



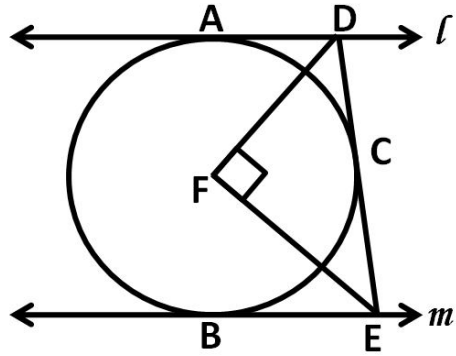
43. In the given figure, a circle touches all the four sides of a quadrilateral ABCD whose sides are AB = 6cm, BC = 7cm and CD = 4cm. Find AD.



44. Prove that "If a line touches a circle and from the point of contact a chord is drawn, the angle which this chord makes with the given line are equal respectively to the angles formed in the corresponding alternate segments."

45. Prove that "If a line is drawn through an end point of a chord of a circle so that the angle formed by it with the chord is equal to the angle subtend by chord in the alternate segment, then the line is a tangent to the circle."

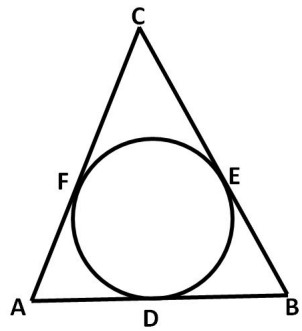
46. In figure. l and m are two parallel tangents at A and B. The tangent at C makes an intercept DE between the tangent l and m. Prove that $\angle DFE = 90^0$



47. In figure, a circle is inscribed in a ΔABC having sides AB = 12 cm, BC = 8cm and AC = 10cm. Find AD, BE and CF.

OR

A circle is inscribed in a ΔABC having sides 8 cm, 10 cm and 12 cm as shown in fig. Find AD, BE and CF.



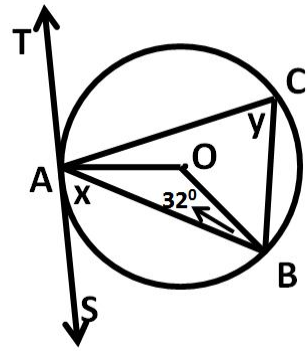
48. If PA and PB are two tangents drawn from a point P to a circle with centre O touching it at A and B, prove that OP is the perpendicular bisector of AB.

49. If ΔABC is isosceles with AB = AC, Prove that the tangent at A to the circumcircle of ΔABC is parallel to BC.

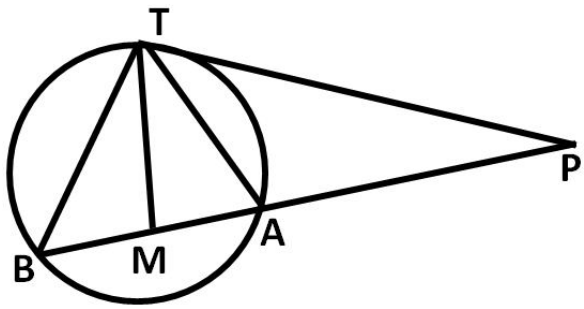
50. Two circles intersect at A and B. From a point P on one of these circles, two lines segments PAC and PBD are drawn intersecting the other circles at C and D respectively. Prove that CD is parallel to the tangent at P.

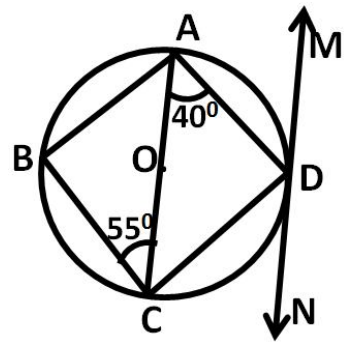
51. Two circles intersect in points P and Q. A secant passing through P intersects the circles at A an B respectively. Tangents to the circles at A and B intersect at T. Prove that A, Q, T and B are concyclic.

52. In the given figure TAS is a tangent to the circle, with centre O, at the point A. If $\angle OBA = 32^0$, find the value of x and y.



53. In the given figure. PT is a tangent and PAB is a secant to a circle. If the bisector of $\angle ATB$ intersect AB in M, Prove that: (i) $\angle PMT = \angle PTM$ (ii) PT = PM





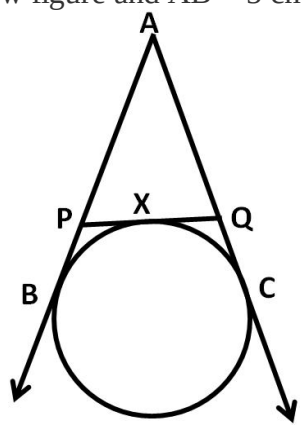
54. In the adjoining figure, ABCD is a cyclic quadrilateral. AC is a diameter of the circle. MN is tangent to the circle at D, $\angle CAD = 40^0, \angle ACB = 55^0$ Determine $\angle ADM$ and $\angle BAD$

55. The diagonals of a parallelogram ABCD intersect at E. Show that the circumcircle of ΔADE and ΔBCE touch each other at E.

56. A circle is drawn with diameter AB interacting the hypotenuse AC of right triangle ABC at the point P. Show that the tangent to the circle at P bisects the side BC.

57. In two concentric circles, prove that all chords of the outer circle which touch the inner circle are of equal length.

58. If AB, AC, PQ are tangents in below figure and AB = 5 cm, find the perimeter of ΔAPQ.



59. If PA and PB are tangents from an outside point P, such that PA = 10 cm and $\angle APB = 60^0$. Find the length of chord AB.

60. From an external point P, tangents PA and PB are drawn to a circle with centre O. If CD is the tangent to the circle at a point E and PA = 14 cm, find the perimeter of ΔPCD.

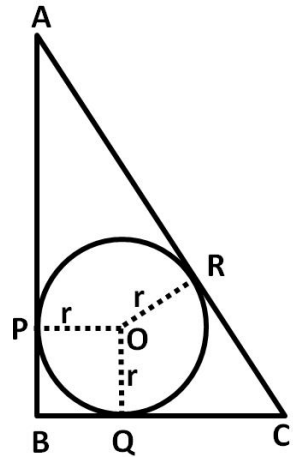
61. Prove that the tangents at the extremities of any chord make equal angles with the chord.

62. From an external point P, two tangents PA and PB are drawn to the circle with centre O. Prove that OP is the perpendicular bisector of AB.

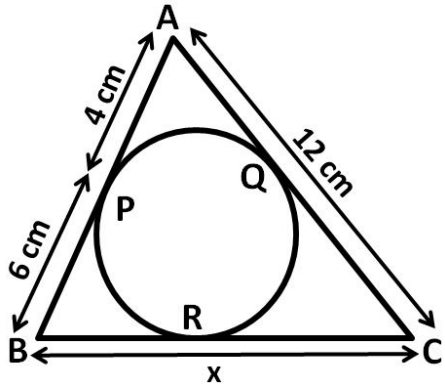
63. The radius of the incircle of a triangle is 4 cm and the segments into which one side divided by the point of contact are 6 cm and 8 cm. Find the other two sides of the triangle.

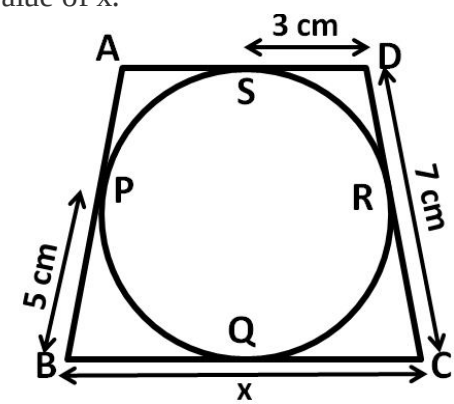
64. From a point P, two tangents PA and PB are drawn to a circle with centre O. If OP = diameter of the circle, show that ΔAPB is an equilateral triangle.

65. In fig. ABC is a right triangle right angled at B such that BC = 6 cm and AB = 8 cm. Find the radius of its incircle.



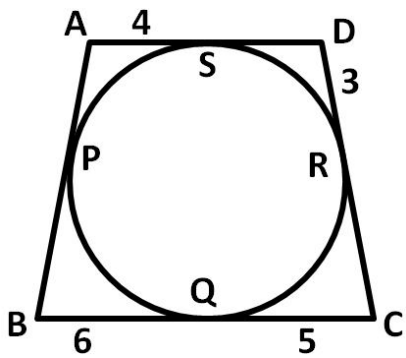
66. In the below figure, ΔABC is circumscribed, find the value of x.

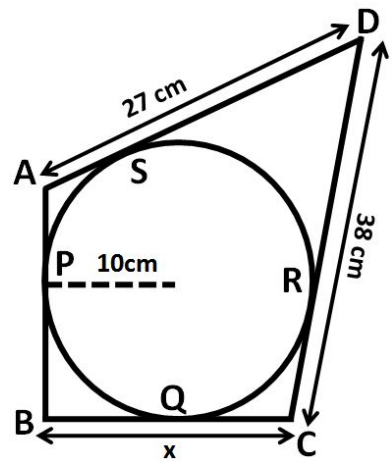




67. In the above right-sided figure, quadrilateral ABCD is circumscribed, find the value of x.

68. In the below figure, quadrilateral ABCD is circumscribed, find the perimeter of quadrilateral ABCD.



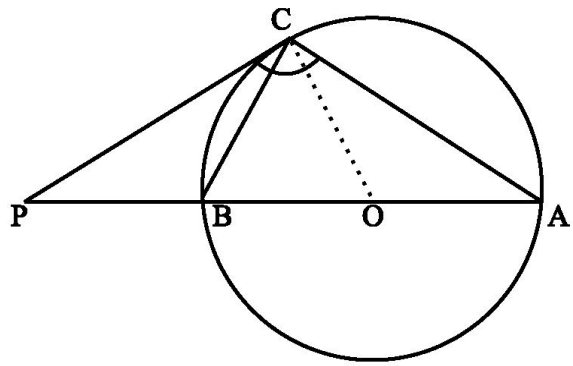


69. In the above right sided figure, quadrilateral ABCD is circumscribed and AD ⊥ DC, find the value of x if the radius of incircle is 10 cm.

70. If an isosceles triangle ABC, in which AB = AC = 6 cm, is inscribed in a circle of radius 9 cm, find the area of the triangle.

71. A is a point at a distance 13 cm from the centre O of a circle of radius 5 cm. AP and AQ are the tangents to the circle at P and Q. If a tangent BC is drawn at a point R lying on the minor arc PQ to intersect AP at B and AQ at C, find the perimeter of the ΔABC.

72. The tangent at a point C of a circle and a diameter AB when extended intersect at P. If $\angle PCA = 110^0$, find $\angle CBA$



73. In a right triangle ABC in which $\angle B = 90°$, a circle is drawn with AB as diameter intersecting the hypotenuse AC at P. Prove that the tangent to the circle at P bisects BC.

74. AB is a diameter and AC is a chord of a circle with centre O such that $\angle BAC = 30°$. The tangent at C intersects extended AB at a point D. Prove that BC = BD.

75. In the below figure from an external point A, tangents AB and AC are drawn to a circle. PQ is a tangent to the circle at X. If AC = 15 cm, find the perimeter of the triangle APQ.

